Next Steps

① Contribute the boundary lemmas to Mathlib.

Goal. Generalize and upstream the auxiliary topological results developed for the trapping region proof.

- `firstExit_mem_frontier` *The first exit time lies on the frontier of the set.*
- `firstExit_mem_closure` *A right-continuous flow maps the first exit time into the closure of the complement.*

These lemmas are completely independent of the Lorenz system and are potentially useful in broader formalizations.

② Extend the formalization of the Lorenz system.

Prove additional fundamental results, including

- Global Existence Theorem
- Existence of an Attracting Set
- Further qualitative properties of the Lorenz flow.

③ Generalize the trapping region lemma.

Develop a version applicable to more general dynamical systems, including

- Higher-dimensional systems
- Non-smooth flows
- Infinite-dimensional systems

$\Longrightarrow$ Towards a reusable formal library for trapping regions and dynamical systems in Lean.